Fundamentals of AI Using Agriculture Data Set

Raksha Case Study – Case 02

Analysing Farmer Queries

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Abstract

This project analyzes farmer queries using Artificial Intelligence techniques. The objective is to transform unstructured farmer questions into meaningful insights that can support agricultural decision-making. The study includes query distribution analysis, intent classification, and similar question retrieval. These techniques help identify farmer concerns, organize queries automatically, and reuse historical knowledge for advisory services.

Objective

• Analyze farmer queries to identify major concerns.
• Classify farmer queries into meaningful categories.
• Retrieve similar historical queries for faster advisory support.
• Demonstrate the role of AI in agricultural decision support systems.

Dataset Description

Dataset Name: raksha-farmer-query.xlsx

The dataset contains farmer queries along with information such as:

• State Name
• District Name
• Crop
• Query Type
• Query Text
• Date Information

Total Records: 2000

Methodology

Step 1: Data Loading and Preprocessing

• Loaded dataset using Pandas.
• Checked dataset structure and missing values.
• Extracted year and month from the date column.

Step 2: Query Distribution Analysis

• Counted queries in each category.
• Created visualizations using bar charts.
• Identified dominant farmer concerns.

Step 3: Intent Classification

• Implemented a rule-based classifier.
• Categorized queries into:

- Weather
- Pest
- Nutrient
- Market
- Government Scheme
- Other

• Compared predicted categories with actual labels.

Step 4: Similar Question Retrieval

• Applied TF-IDF Vectorization.
• Used Cosine Similarity.
• Retrieved top 5 similar historical queries.

Flowchart

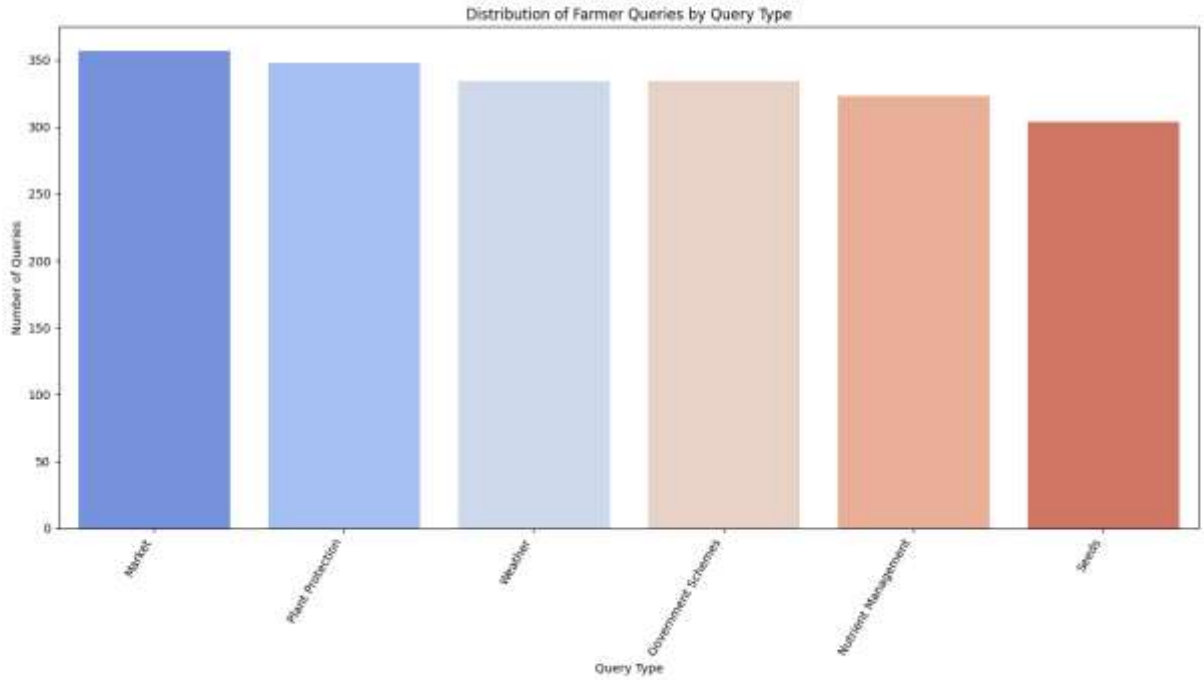
Start
↓
Load Dataset
↓
Data Preprocessing
↓
Query Distribution Analysis
↓
Intent Classification
↓
Similar Query Retrieval
↓
Generate Insights
↓
End

Task 1: Query Distribution Analysis

The analysis showed the distribution of farmer concerns across different categories. Weather, pest management, nutrient management, market information, and government schemes were among the major topics discussed by farmers.

Inference

• Farmers frequently seek guidance related to crop management.
• Query patterns help identify emerging agricultural risks.
• Such analysis can support agricultural planning.



Task 2: Intent Classification

A rule-based classifier was used to automatically categorize farmer queries.

Inference

• AI can organize large volumes of agricultural queries.
• Classification helps route queries to appropriate experts.
• Keyword-based classification is simple but has limitations.

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Total queries: 2000
Correctly classified queries: 2000
Misclassified queries: 0
Accuracy of rule-based classifier: 100.00%
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Task 3: Similar Question Retrieval

TF-IDF and cosine similarity were used to retrieve similar farmer queries.

Inference

• Historical knowledge can be reused effectively.
• Similar query retrieval supports FAQ systems.
• It reduces response time for agricultural helpdesks.

1 to 5 of 5 entries Filter

index	QueryText	Category
0	How to make pasta?	Cooking
1	Best pasta recipes	Cooking
2	Italian pasta dishes	Cooking
3	Quick dinner ideas	Food
4	Healthy breakfast options	Food

Conclusion

This project successfully demonstrated the application of Artificial Intelligence in analyzing farmer queries. Query analysis, intent classification, and similar question retrieval transformed raw text data into useful insights. The study highlights the potential of AI-driven systems to improve agricultural advisory services, reduce response times, and support farmers in making informed decisions.

What Did This Case Study Help Me Infer?

This case study helped me understand how Artificial Intelligence can be applied to real-world agricultural problems. By analyzing farmer queries, I learned that large amounts of unstructured text data can be converted into meaningful information. The analysis revealed the major concerns of farmers, such as weather conditions, crop protection, nutrient management, market-related issues, and government schemes.

The intent classification task demonstrated how AI can automatically categorize farmer queries, making it easier to organize and process large volumes of information. The similar question retrieval task showed how historical knowledge can be reused to provide faster and more consistent responses to farmers.

Overall, this case study highlighted the importance of AI in improving agricultural advisory systems, reducing response time, and supporting better decision-making for farmers.

Blog Article Summary

Agriculture is one of the most important sectors of the economy, and farmers frequently seek guidance regarding weather, pests, nutrients, markets, and government schemes. However, handling thousands of farmer queries manually can be time-consuming and inefficient. This case study explored how Artificial Intelligence can be used to analyze and organize farmer queries effectively.

The first step involved analyzing the distribution of queries to identify the most common concerns among farmers. This provided valuable insights into agricultural challenges and emerging risk areas. The second step focused on intent classification, where farmer queries were automatically categorized into predefined groups such as weather, pest management, nutrient management, market-related queries, and government schemes.

The third task involved retrieving similar past queries using text similarity techniques. This approach can help agricultural helpdesks provide faster responses by reusing previously answered questions. It can also support the development of FAQ systems and intelligent advisory platforms.

This project demonstrates how AI can transform raw agricultural data into actionable insights, helping farmers make informed decisions and improving overall farm productivity.

Can This Be a GitHub Repository?

Yes.

The project is well-suited for a GitHub repository because it contains:

- Dataset analysis using Python and Google Colab
- Intent classification model
- Similar query retrieval system
- Visualizations and insights
- Project report and documentation

A GitHub repository allows students, researchers, and developers to learn from the project, contribute improvements, and collaborate on agricultural AI solutions.

Better Solution and Future Scope

- Use Machine Learning algorithms for better classification accuracy.
- Apply Transformer-based models such as BERT for understanding query context.
- Support multiple Indian languages used by farmers.
- Build a chatbot for real-time farmer assistance.
- Create a web dashboard for query monitoring and analytics.
- Integrate weather forecasts and market price data for personalized recommendations.

Final Thoughts

This case study is not only about completing an assignment but also about understanding how technology can solve real-world problems. It inspired me to think about practical applications of AI in agriculture and how innovative solutions can directly benefit farmers. Such projects encourage creativity, problem-solving, and the development of technology that creates a positive social impact.

References

1. Raksha Farmer Query Dataset
2. Python Pandas Documentation
3. Scikit-Learn Documentation
4. Google Colab Platform
5. Fundamentals of AI Using Agriculture Data Set Course Material

Output Screenshots

1. Query Distribution Bar Chart
2. Intent Classification Accuracy Output
3. Similar Question Retrieval Output